

Smoothing & Nonparametric Methods

Kernel Smoothing, Splines, and Local Regression

StatHarbor Analytics

Advanced Analytics Series

Course Roadmap

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Why Smoothing?

The Parametric Starting Point

In many statistical models, we assume a finite-dimensional form for the mean structure:

$$Y = f(X) + \varepsilon, \quad \mathbb{E}[\varepsilon \mid X] = 0.$$

Example: linear regression

A classical linear model assumes

$$f(X) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p.$$

The estimation problem is reduced to a finite set of coefficients.

- This is efficient when the assumed form is close to the truth.
- It can be badly biased when the functional form is misspecified.

Why Parametric Models Can Fail

When the true regression surface is nonlinear, rigid functional assumptions can distort inference.

- Thresholds, saturation, curvature, and local features may be missed.
- Adding global polynomial terms can help, but often creates instability.
- A local change in one region can alter the fit far away.

Pragmatic lesson

The issue is not whether parametric models are “good” or “bad.” The issue is whether the assumed structure is credible for the scientific problem.

What “Nonparametric” Means

Definition

Nonparametric methods do *not* mean “parameter-free.” They avoid imposing a fixed low-dimensional functional form for the target curve in advance.

- The effective complexity of the fitted function may grow with sample size.
- Structure is introduced through smoothness, locality, or penalties rather than a fixed algebraic form.
- The goal is flexibility without abandoning statistical discipline.

Bias–Variance Tradeoff

For point estimation of a regression function at x_0 ,

$$\mathbb{E}\left[(\hat{f}(x_0) - f(x_0))^2\right] = \text{Var}(\hat{f}(x_0)) + \left(\text{Bias}(\hat{f}(x_0))\right)^2.$$

Under the additive-noise model, prediction error adds irreducible noise:

$$\mathbb{E}\left[(Y - \hat{f}(x_0))^2 \mid X = x_0\right] = \text{Var}(\varepsilon \mid X = x_0) + \text{Var}(\hat{f}(x_0)) + \left(\text{Bias}(\hat{f}(x_0))\right)^2.$$

- More flexibility usually lowers bias.
- More flexibility usually raises variance.

Kernel Density Estimation

From Histograms to Smooth Density Estimation

Before smoothing $\mathbb{E}[Y | X]$, start with a simpler target: the density $f(x)$ of a univariate variable.

- Histograms estimate density by binning observations.
- They are easy to interpret but discontinuous and sensitive to bin boundaries.

Motivation

Kernel density estimation replaces hard bins with smooth local weighting.

A Naive Local Density Estimator

A simple local estimator counts observations in a window around x :

$$\hat{f}(x) = \frac{1}{2nh} \sum_{i=1}^n I(x - h < X_i < x + h).$$

- h controls window width.
- This removes fixed bin edges.
- But the estimate is still discontinuous because observations enter and leave the window abruptly.

Kernel density estimation smooths the local count:

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{X_i - x}{h}\right).$$

Interpretation

Each observation contributes a local bump centered at X_i . The density estimate is the sum of those bumps.

- $K(\cdot)$: kernel function
- h : bandwidth, the main tuning parameter

Common Kernels

- **Gaussian:**

$$K(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}$$

- **Epanechnikov:**

$$K(u) = \frac{3}{4}(1 - u^2), \quad |u| \leq 1$$

- **Uniform:**

$$K(u) = \frac{1}{2}, \quad |u| \leq 1$$

- **Triangular:**

$$K(u) = 1 - |u|, \quad |u| \leq 1$$

Important

In practice, bandwidth matters far more than kernel choice.

Bandwidth Controls the Smoothness

Small h

- very local fit
- low bias
- high variance
- noisy estimate

Large h

- broader averaging
- higher bias
- lower variance
- oversmoothed estimate

Core lesson

Bandwidth is the central bias–variance tuning parameter in kernel methods.

Kernel Regression and Local Regression

From Density Estimation to Regression

Now the target is the regression function

$$f(x_0) = \mathbb{E}[Y \mid X = x_0].$$

A generic smoother estimates $f(x_0)$ using nearby responses:

$$\hat{f}(x_0) = \sum_{i=1}^n w_i(x_0) Y_i, \quad \sum_{i=1}^n w_i(x_0) = 1.$$

Smoothing principle

Observations with predictor values closer to x_0 receive larger weights.

The Nadaraya–Watson estimator uses kernel weights:

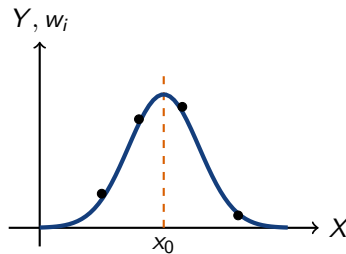
$$\hat{f}_{NW}(x_0) = \frac{\sum_{i=1}^n K\left(\frac{X_i - x_0}{h}\right) Y_i}{\sum_{i=1}^n K\left(\frac{X_i - x_0}{h}\right)}.$$

- It is a local constant estimator.
- It computes a weighted average of nearby Y_i values.
- Simple and intuitive, but not always stable at the boundaries.

Visualizing Kernel Regression

At a target point x_0 :

1. Center a kernel at x_0 .
2. Evaluate its weight at each observed X_i .
3. Give larger influence to nearby responses.
4. Normalize to obtain a local average.



Nadaraya–Watson performs poorly near the edges of the support.

- At the boundary, the kernel neighborhood is asymmetric.
- There are observations on one side but not the other.
- A local constant fit is then pulled inward.

Consequence

The estimator tends to flatten near boundaries and can be noticeably biased.

Local Linear Regression

A better local model assumes the regression function is approximately linear near x_0 :

$$\min_{\beta_0, \beta_1} \sum_{i=1}^n K\left(\frac{X_i - x_0}{h}\right) [Y_i - \beta_0 - \beta_1(X_i - x_0)]^2.$$

The fitted value at x_0 is the local intercept:

$$\hat{f}_{LL}(x_0) = \hat{\beta}_0.$$

Why it matters

Local linear regression substantially reduces boundary bias while retaining local adaptivity.

Local Polynomial Regression

More generally, fit a polynomial of degree p locally:

$$\min_{\beta} \sum_{i=1}^n K\left(\frac{X_i - x_0}{h}\right) \left[Y_i - \sum_{j=0}^p \beta_j (X_i - x_0)^j \right]^2.$$

- $p = 0$: local constant
- $p = 1$: local linear
- $p = 2$: local quadratic

Practice

Local linear regression is often the default because it offers a strong bias–variance compromise.

LOESS is a practical implementation of local polynomial regression.

- Usually uses a nearest-neighbor bandwidth rather than a fixed h .
- The user specifies a span α , such as 0.3.
- The effective window widens in sparse regions and narrows in dense regions.
- A tricube weight function is commonly used.

Use case

LOESS is especially effective for exploratory smoothing of scatterplots.

LOESS can be made less sensitive to outliers by iteratively reweighting observations.

1. Fit an initial LOESS curve.
2. Compute residuals.
3. Downweight observations with unusually large residuals.
4. Refit using both proximity weights and robustness weights.

Interpretation

This protects the local fit from being dominated by a few extreme points.

Bandwidth Selection

Bandwidth Selection Is the Main Tuning Problem

For kernel methods, kernel choice is usually secondary. Bandwidth choice is decisive.

Question

How much local averaging should we allow?

Two broad strategies are common:

- asymptotic plug-in rules
- data-adaptive criteria such as cross-validation

For KDE, the asymptotic mean integrated squared error has the form

$$\text{AMISE}(h) \approx \frac{R(K)}{nh} + \frac{1}{4}h^4\mu_2(K)^2R(f'').$$

- the first term is variance-dominated
- the second term is squared-bias-dominated

Implication

The optimal bandwidth balances a variance term that decreases with h against a bias term that increases with h .

Minimizing the AMISE yields the familiar bandwidth rate

$$h_{\text{opt}} \propto n^{-1/5}.$$

- As sample size grows, bandwidth should shrink.
- It shrinks slowly, reflecting the difficulty of nonparametric estimation.

Key comparison

Nonparametric estimators generally converge more slowly than well-specified parametric estimators.

A practical way to select bandwidth in regression is leave-one-out cross-validation:

$$\text{CV}(h) = \sum_{i=1}^n \left(Y_i - \hat{f}_{-i}(X_i; h) \right)^2.$$

- $\hat{f}_{-i}$ is the smoother fit without observation i .
- Leaving i out prevents trivial interpolation.
- The bandwidth minimizing CV is often a good predictive choice.

For linear smoothers, $\hat{Y} = SY$, GCV approximates leave-one-out CV without refitting n times:

$$\text{GCV}(h) = \frac{\frac{1}{n} \sum_{i=1}^n \left(Y_i - \hat{f}(X_i) \right)^2}{\left(1 - \frac{1}{n} \text{tr}(S) \right)^2}.$$

Why it is useful

GCV is fast, stable, and central in smoothing splines and many spline-based methods.

A classic rule of thumb for KDE with a Gaussian kernel is

$$\hat{h}_{\text{ROT}} \approx 1.06 \hat{\sigma} n^{-1/5}.$$

- fast
- simple
- often adequate for roughly unimodal symmetric densities

Limitation

Rules of thumb can fail badly for skewed, multimodal, or heavy-tailed distributions.

Regression Splines

Kernel methods build a local fit at each target point. Splines instead build a global function from local pieces.

Idea

Partition the predictor axis using knots

$$\xi_1, \dots, \xi_K$$

and fit piecewise polynomial functions joined smoothly at those knots.

From Piecewise Linear Fits to Splines

With a knot at c , separate linear fits on either side produce a discontinuity unless constraints are imposed.

A convenient basis term is

$$(X - c)_+ = \begin{cases} X - c, & X > c, \\ 0, & X \leq c. \end{cases}$$

Then

$$f(X) = \beta_0 + \beta_1 X + \beta_2 (X - c)_+$$

is continuous at c .

Interpretation

The basis function allows the slope to change after the knot while preserving continuity.

Cubic splines use cubic polynomial pieces joined at knots with continuity constraints.

- the function is continuous
- the first derivative is continuous
- the second derivative is continuous

This yields a smooth curve without sharp corners.

A cubic spline with knots $\xi_1, \dots, \xi_K$ can be written as

$$f(X) = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \sum_{k=1}^K \theta_k (X - \xi_k)_+^3.$$

- flexible and conceptually transparent
- can be estimated by least squares once the basis is constructed

Natural Cubic Splines

Standard cubic splines can behave erratically near the boundaries.

Natural cubic spline constraint

Force the spline to be linear beyond the boundary knots.

- stabilizes edge behavior
- reduces sensitivity to extrapolation
- often preferred in applied regression

For regression splines, the analyst must choose:

1. the number of knots
2. the knot locations
 - A common default is to place knots at quantiles of X .
 - Selection can be guided by cross-validation or information criteria.

Why B-Splines Are Preferred in Software

The truncated power basis is intuitive but can be numerically unstable.

B-splines

B-splines span the same spline space using basis functions with local support.

- improved numerical conditioning
- sparse design matrices
- better computational stability

Smoothing Splines

Regression splines require explicit knot selection. Smoothing splines take a different route.

Idea

Place a knot at every distinct predictor value and control complexity with a roughness penalty rather than by limiting knots directly.

A smoothing spline minimizes

$$\sum_{i=1}^n (Y_i - f(X_i))^2 + \lambda \int [f''(t)]^2 dt.$$

- first term: fidelity to the data
- second term: penalty on curvature
- λ : smoothing parameter

Interpretation

Large λ enforces smoothness. Small λ allows greater flexibility.

Why Penalize the Second Derivative?

The second derivative measures curvature.

- Straight lines have $f''(t) = 0$.
- Highly wiggly functions have large curvature.
- Penalizing $\int [f''(t)]^2 dt$ discourages unnecessary oscillation.

Takeaway

The penalty does not forbid nonlinearity. It discourages rough nonlinearity unsupported by the data.

Theorem

The minimizer of the penalized least squares criterion over a suitable function class is a natural cubic spline with knots at the observed predictor values.

This converts an infinite-dimensional optimization problem into a finite-dimensional spline problem.

If $f(x) = \sum_{j=1}^N N_j(x)\theta_j$, then the criterion becomes

$$(Y - N\theta)^\top (Y - N\theta) + \lambda \theta^\top \Omega_N \theta,$$

where

$$(\Omega_N)_{jk} = \int N_j''(t) N_k''(t) dt.$$

The solution is

$$\hat{\theta} = (N^\top N + \lambda \Omega_N)^{-1} N^\top Y.$$

Connection

This is a penalized least squares problem analogous to ridge regression.

Smoothing splines are often summarized through effective degrees of freedom:

$$df_{\lambda} = \text{tr}(S_{\lambda}).$$

- df_{λ} measures model flexibility
- as $\lambda \rightarrow \infty$, the fit approaches a straight line
- as $\lambda \rightarrow 0$, the fit approaches interpolation

A common data-driven criterion is

$$\text{GCV}(\lambda) = \frac{\frac{1}{n} \sum_{i=1}^n \left(Y_i - \hat{f}_\lambda(X_i) \right)^2}{\left(1 - \frac{df_\lambda}{n} \right)^2}.$$

- widely used in smoothing spline software
- balances fit against effective complexity

Every linear smoother can be written as

$$\hat{f}(x_0) = \sum_{i=1}^n w_i(x_0) Y_i.$$

Smoothing splines therefore admit an implicit weighting representation.

Interpretation

Although derived through penalized splines, they still behave like adaptive local smoothers.

Higher Dimensions and GAMs

The Curse of Dimensionality

Local methods depend on nearby observations. In high dimensions, neighborhoods become sparse.

- data spread out rapidly as dimension increases
- local averaging requires increasingly large neighborhoods
- the notion of “local” becomes much less informative

Consequence

Naive multivariate smoothing breaks down quickly as dimensionality grows.

A major practical compromise is the additive model:

$$\mathbb{E}[Y \mid X] = \beta_0 + f_1(X_1) + \cdots + f_p(X_p).$$

- each component is estimated as a one-dimensional smoother
- nonlinear effects are retained
- the dimensionality problem is dramatically reduced

Backfitting estimates additive components iteratively.

1. initialize $\hat{\beta}_0$ and the smooth components
2. compute partial residuals for one component at a time
3. smooth those residuals against the corresponding predictor
4. cycle until convergence

Practical value

Backfitting allows flexible additive structure without fitting a full high-dimensional surface directly.

GAMs extend additive models beyond Gaussian outcomes:

$$g(\mathbb{E}[Y \mid X]) = \beta_0 + f_1(X_1) + \cdots + f_p(X_p).$$

- logistic GAMs for binary outcomes
- Poisson GAMs for counts
- other link-function settings for skewed outcomes

Why GAMs matter

They combine flexibility, interpretability, and compatibility with standard regression frameworks.

Interpreting GAM Components

Because the model is additive, each fitted smooth $\hat{f}_j(X_j)$ can be plotted separately.

These component plots reveal:

- approximate linearity
- threshold behavior
- saturation
- U-shaped or plateau effects

Important distinction

These are component-effect plots, not generic black-box feature importance summaries.

Extensions

Two useful extensions are:

- **MARS:** piecewise linear basis expansion with automatic knot and interaction search
- **Projection Pursuit Regression:**

$$f(X) = \sum_{m=1}^M g_m(w_m^\top X)$$

which smooths low-dimensional projections rather than raw coordinates

These methods relax strict additivity while remaining more structured than unrestricted black-box learners.

Smoothing methods are historically and conceptually important for modern ML.

- kernel ideas underlie many nonlinear similarity-based methods
- spline bases remain important in modern function approximation
- additive structure reappears in interpretable ML
- projection-based smoothers anticipate neural-network-style representations

Caution

These are conceptual links, not exact equivalences.

Conclusion

Which Smoother for Which Task?

- **Exploratory visualization:** LOESS
- **Density estimation:** KDE
- **Stable smooth regression in 1D:** smoothing splines
- **Moderate flexibility with explicit basis control:** regression splines
- **Multiple predictors with interpretability:** GAMs

Final Takeaways

1. Nonparametric methods replace rigid global form assumptions with smoothness or locality assumptions.
2. Flexibility must be controlled through bandwidths, penalties, or basis complexity.
3. Bandwidth and smoothing-parameter selection are central to performance.
4. In higher dimensions, additive or other structural assumptions often become necessary.

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